

ETHAN STERLING

iethansterling@gmail.com | github.com/EthanSterling04 | [linkedin.com/in/ethan-sterling-2004](https://www.linkedin.com/in/ethan-sterling-2004) | <https://ethansterling.com>

EDUCATION

Northwestern University, Evanston, IL

September 2022 – June 2026

M.S. in Computer Science and B.A. in Economics, Minor in Data Science

Graduate GPA: 3.9/4.0

- Awards: Departmental Honors in Economics, Phi Beta Kappa, Dean's List, Innovation Fellowship
- Undergraduate Teaching Assistant for Introduction to Computer Systems
- Coursework: Kernel & Low Level Software Development, Operating Systems, Networking, System Security, Programming Languages, Data Structures & Algorithms, Mechanism Design, Linear Algebra, Machine Learning

PROFESSIONAL EXPERIENCE

Federal Reserve Bank of Chicago

Chicago, IL

Software Engineering Intern

June 2024 – August 2024 | June 2025 – August 2025

- Architected RESTful APIs using Spring Boot to replace production database links, aiding cloud migration effort
- Created dynamic Spring Batch pipeline for high-performance, parallel data migration from Oracle to PostgreSQL
- Implemented high-availability disaster recovery solution on AWS using Terraform to deploy multi-region application on ECS Fargate, enabling automated, health-check-driven failover managed by Route 53, Lambda, and CloudWatch
- Engineered self-healing DevSecOps pipeline with GitLab CI to automate OSS dependency updates in Spring/Angular
- Leveraged Docker, Terraform, Mockito unit tests, and built GitLab CI/CD pipelines for deployment and scalability

Mayfest Productions

Evanston, IL

Co-Chair (2025–26) | Executive Producer (2024–25) | Producer (2022–24)

November 2022 – June 2026

- Led 115-person organization and 10 committees to produce \$1M, 12,000-attendee music festival, adding a third stage
- Drove \$150K+ in sponsorship and wristband revenue while cutting \$10K in production costs via vendor consolidation
- Handled University safety, compliance, and contracts for 16+ artist teams, re-booking dropouts within 24 hours

Fulminare Holdings

San Diego, CA

Software Engineering Intern

June 2023 – September 2023

- Shipped Swift data import tools for seamless onboarding plus an OpenAI Twitter bot driving 8,000+ new users

RESEARCH

Prescience Lab – The Buoyancy Project (FPVM)

Evanston, IL

Research Assistant

January 2025 – June 2026

- Built a JIT compiler in C, translating a stack-based floating-point IR into x86-64 assembly, leveraging compiled-code caching to achieve a 3.5x average speedup over interpretation and match original architecture-specific emulator
- Abstracted hardware dependencies by lowering same IR to LLVM IR, ensuring full architecture independence
- Enhanced VM instruction set with x86 comparison, precisely manipulating EFLAGS for complex conditional logic
- Defined architectural roadmap for 256-bit AVX vector instruction support to enhance parallel processing capabilities

Prescience Lab – The Constellation Project (DMPL)

Evanston, IL

Research Assistant

April 2024 – September 2024

- Scaled functional programming language MaPLe to a distributed memory environment using MPI and C
- Developed efficient parallel garbage collection for MPL, scaling to hundreds of cores and 1TB+ heap sizes
- Optimized whole-program compilation with MLton for performance competitive with C/C++

PROJECTS

Limit Order Book and Matching Engine (C++20)

September 2026 – Present

- Built a price-time-priority limit order book and matching engine in C++20 with an arena allocator and intrusive levels
- Benchmarked swappable price-level indexes (map vs sorted vector vs dense array), cutting p99 add latency to 110 ns
- Verified correctness with 15 scenario tests and a 100k-event property test asserting five invariants after every event

Honors Thesis: Market Concentration as a Signal for Tail Risk

January 2026 – May 2026

- Built a 2005–2026 monthly panel of 11 sector ETFs in Python from Bloomberg, OptionMetrics, BEA I-O, and FRED
- Designed a novel Input-Output-weighted measure of upstream concentration to predict options-implied tail risk
- Ran two-way fixed-effects predictive regressions; validated with Driscoll–Kraay SEs and out-of-sample tests

TECHNICAL SKILLS

Programming Languages: Java, C, C++, Python, JavaScript, SQL, STATA, Swift, C#, R, Assembly, LISP

Tools and Technologies: Git, Unix shell, GDB, Valgrind, LLVM, Docker, Pandas, NumPy, Matplotlib, AWS, PostgreSQL, scikit-learn, TensorFlow, Jupyter, Spring Boot, Jenkins, Terraform, Mockito, GitLab, Vim, React, Angular, Hibernate, H2